Answer key. Answers are in magenta.

NRES 746 – Week 2, Lecture 2: Limits and a Derivative

Instructions: Work alone for about 5 minutes. Then compare with a neighbor.

Part 1: What happens at the edges?

For each function below, use the "throw out the lower-order terms" trick to state what happens as $x \to 0$ and as $x \to \infty$. You don't need to show work, just the two limiting behaviors. We've already done the first two together in class!

1a. Michaelis-Menten: $f(x) = \dfrac{ax}{b+x}$

As $x \to 0$: $\approx (a/b)\,x \to 0$ As $x \to \infty$: $\to a$

1b. Ricker: $f(x) = ax\,e^{-bx}$

As $x \to 0$: $\approx ax \to 0$ As $x \to \infty$: $\to 0$

1c. Logistic: $f(x) = \dfrac{1}{1+e^{-bx}}$

As $x \to 0$: $\to 1/2$ As $x \to \infty$: $\to 1$

1d. Holling type III: $f(x) = \dfrac{ax^2}{b^2+x^2}$

As $x \to 0$: $\approx (a/b^2)\,x^2 \to 0$ As $x \to \infty$: $\to a$

Part 2: What kind of point is $x = 0$?

The logistic function, $f(x) = \dfrac{1}{1+e^{-bx}}$, is the simplest sigmoid in the bestiary. It is symmetric around $x = 0$.

2a. Qualitatively, how would you characterize the derivative of the logistic function at x=0 relative to the derivative of this function at all other inputs (x)?

It's the steepest point: the derivative is largest at $x = 0$ and shrinks toward 0 in both directions.

2b. In Lab 1 you found that the Ricker function's derivative is zero at its peak. This makes it a 'critical point'. Is $f'(0)$ also zero at the logistic's midpoint (x=0)? If not, what kind of special point is $x = 0$ instead? (Hint: think about the curve's concavity just before and just after $x = 0$ – what would that imply about the *second* derivative at $x = 0$?)

No: $f'(0) = b/4 > 0$. It's an inflection point. The curve is concave up before 0 and concave down after, so $f''(0) = 0$.

Optional, for fun! Rewrite the logistic function $f(x)$ as $(1+e^{-bx})^{-1}$, then find $f'(x)$ using the chain rule (twice: once for the outer power, once for the inner exponential).

Reminders: $\dfrac{d}{dx}\,u^n = n\,u^{n-1}\,u'$ $\dfrac{d}{dx}\,e^{g(x)} = g'(x)\,e^{g(x)}$

Show your work:

$$f'(x) = -1\,(1+e^{-bx})^{-2} \cdot (-b\,e^{-bx}) = \frac{b\,e^{-bx}}{(1+e^{-bx})^2}$$

Now evaluate $f'(x)$ at $x = 0$.

$$f'(0) = \frac{b \cdot 1}{(1+1)^2} = b/4$$